S1 VALIDATION SET CONSTRUCTION

TitanVul sampling. We draw the 150 TitanVul pairs from its 17,101 C/C++ pairs with a fixed random seed (42), sampling by CWE in proportion to each CWE's frequency:

- **Proportional share.** Each CWE receives a share of the 150 pairs in proportion to how often it appears in TitanVul C/C++.
- **Minimum of one.** Every CWE receives at least one pair, so rare CWEs are slightly overrepresented. If this pushes the total over budget, pairs are removed at random until the budget is met.
- **Unlabeled pairs.** Pairs without a usable CWE (a blank label, `NVD-CWE-noinfo` or `NVD-CWE-Other`) are sampled as their own group, also in proportion to their frequency, with a minimum of 5. This group contributes 84 pairs: 79 with a blank label and 5 with only an NVD placeholder.

CWE distribution. Table S1.1 lists every CWE in the validation set D and the held-out PrimeVul set.

- **TitanVul (150 pairs).** 79 pairs have no CWE label. The other 71 are labeled and cover 58 distinct CWEs; 26 of them carry several CWEs, such as `CWE-119,CWE-190,CWE-787`. Eight pairs include an NVD placeholder instead of, or alongside, a real CWE: `NVD-CWE-Other` (5) and `NVD-CWE-noinfo` (3). Five of these have no real CWE. The rest form a long tail: CWE-345, CWE-400 and CWE-787 have 3 pairs each, and most other CWEs appear once.
- **CWEval (52 pairs).** Every pair has exactly one CWE, covering 29 distinct CWEs.
- **Overall.** 123 of the 202 pairs have a CWE label, and D covers 72 distinct CWEs. CWEval leans toward web and cryptographic weaknesses, while TitanVul leans toward memory-safety bugs.
- **PrimeVul held-out set (430 pairs).** Every pair has exactly one CWE, covering 61 distinct CWEs, 31 of which also appear in D. Memory-safety bugs dominate: CWE-787 (70), CWE-125 (47), CWE-703 (47), CWE-476 (39) and CWE-416 (29) account for 232 of the 430 pairs, and 25 CWEs appear only once.

Table S1.1: All 102 CWEs in the validation set D (TitanVul and CWEval, 72 CWEs) and the held-out PrimeVul set (430 pairs, 61 CWEs), with the number of pairs from each source. 31 CWEs appear in both sets. A TitanVul pair with several CWEs counts once for each. Not listed: 79 TitanVul pairs with no CWE label, and the NVD placeholders `NVD-CWE-Other` (5 TitanVul pairs) and `NVD-CWE-noinfo` (3).

CWE	TitanVul	CWEval	PrimeVul
CWE-16	1	0	0
CWE-17	0	0	1
CWE-19	1	0	0
CWE-20	4	2	14
CWE-22	0	4	6
CWE-59	1	0	2
CWE-77	1	0	0
CWE-78	1	2	0
CWE-79	1	2	2
CWE-94	0	0	2
CWE-113	0	2	0
CWE-117	0	2	0
CWE-119	6	1	14
CWE-120	4	1	7
CWE-121	1	0	0
CWE-122	2	0	3

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Table S1.1 continued

CWE	TitanVul	CWEval	PrimeVul
CWE-125	6	1	47
CWE-129	1	0	0
CWE-131	0	1	0
CWE-134	1	1	1
CWE-170	0	1	0
CWE-189	1	0	5
CWE-190	2	1	11
CWE-191	1	0	2
CWE-193	0	0	2
CWE-200	1	0	16
CWE-212	1	0	1
CWE-252	0	0	1
CWE-255	1	0	0
CWE-264	0	0	1
CWE-269	1	0	4
CWE-275	0	0	1
CWE-276	1	0	2
CWE-284	0	0	4
CWE-287	1	0	1
CWE-288	0	0	1
CWE-295	2	0	1
CWE-310	0	0	2
CWE-311	1	0	0
CWE-320	1	0	0
CWE-326	0	4	0
CWE-327	1	4	1
CWE-329	0	2	0
CWE-331	1	0	0
CWE-345	3	0	0
CWE-346	1	0	0
CWE-347	2	2	0
CWE-354	0	0	2
CWE-362	5	0	8
CWE-369	1	0	14
CWE-377	0	1	0
CWE-388	1	0	0
CWE-391	1	0	0
CWE-399	0	0	3
CWE-400	3	1	3
CWE-401	0	0	8
CWE-415	0	0	9
CWE-416	1	0	29
CWE-434	0	0	1
CWE-436	1	0	0
CWE-444	0	0	1
CWE-459	1	0	0
CWE-468	0	1	0
CWE-476	4	1	39
CWE-494	1	0	0
CWE-522	0	0	1
CWE-552	0	0	1
CWE-601	1	0	0
CWE-617	0	0	12
CWE-643	0	2	0
CWE-665	0	0	1
CWE-667	1	0	0
CWE-668	2	0	1
CWE-672	1	0	1
CWE-676	0	1	0
CWE-693	1	0	0
CWE-703	0	0	47

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Table S1.1 continued

CWE	TitanVul	CWEval	PrimeVul
CWE-704	0	0	2
CWE-732	1	2	1
CWE-754	0	0	2
CWE-755	1	0	0
CWE-760	0	2	0
CWE-770	0	0	1
CWE-772	0	0	2
CWE-787	3	1	70
CWE-788	1	0	0
CWE-822	1	0	0
CWE-824	0	0	1
CWE-825	1	0	0
CWE-834	0	0	1
CWE-835	0	0	5
CWE-840	1	0	0
CWE-843	0	0	3
CWE-862	1	0	1
CWE-863	1	0	1
CWE-908	0	0	2
CWE-909	1	0	1
CWE-918	0	4	0
CWE-924	0	0	1
CWE-943	1	2	0
CWE-1284	1	0	0
CWE-1333	0	1	0